SQL Quick Reference

Basic SQL Commands

CREATE TABLE / Define structure

```
CREATE TABLE customers (
  id INTEGER PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  email VARCHAR(255),
  balance DECIMAL(10,2) DEFAULT 0
);
```

INSERT / Add records

```
INSERT INTO customers (id, name,
  email, balance)
VALUES (1, 'Alice',
  'alice@example.com', 100.00);
```

SELECT / Read records

```
SELECT * FROM customers;
SELECT name, email FROM customers
  WHERE balance > 100;
SELECT * FROM customers
  WHERE balance > 100 AND name = 'Bob';
SELECT * FROM customers
  WHERE balance < 50 OR name = 'Alice';
SELECT * FROM customers
  ORDER BY balance DESC LIMIT 5;
```

UPDATE / Modify records

```
UPDATE customers SET balance = 500
  WHERE id = 1;
UPDATE customers
  SET balance = balance + 50
  WHERE id = 1;
```

DELETE / Remove records

```
DELETE FROM customers WHERE id = 1;
DELETE FROM customers;
```

Without WHERE, removes every row.

DROP TABLE / Remove table

```
DROP TABLE IF EXISTS customers;
```

Removes both structure and data.

WHERE / Common conditions

```
id = 1             -- equal
id <> 1            -- not equal
balance >= 100     -- > < >= <=
name LIKE 'A%'     -- starts with A
id IN (1, 2, 3)    -- matches a set
balance BETWEEN 50 AND 100
email IS NULL      -- missing value
balance > 0 AND id < 10
  -- also use OR to combine tests
```

Remember: UPDATE or DELETE without WHERE affects every row.

For user-provided values, use parameterized queries rather than string concatenation.